PRACTICAL 4:

TO STUDY SOFTWARE ESTIMATION TECHNIQUES AND THEN CALCULATE THE ESTIMATED EFFORT, DEVELOPMENT TIME, AND COST FOR CASE STUDY

Software Estimation Techniques

Software estimation is used to predict the size, effort, time, and cost required to develop a software project.

Common techniques are:

1. **LOC (Lines of Code)**
   - Estimates software size based on the number of lines of source code.
   - Example: 5,000 LOC.
2. **Function Point (FP) Analysis**
   - Estimates size based on the functionality provided to users.
   - It considers:
     - External Inputs (EI)
     - External Outputs (EO)
     - External Queries (EQ)
     - Internal Logical Files (ILF)
     - External Interface Files (EIF)
3. **COCOMO Model**
   - Estimates **effort, development time, and cost** from the estimated LOC.
   - Basic COCOMO formulas:

   Effort = a(KLOC)^b

   Development Time = c(Effort)^d

   Cost = Effort * Cost per Person Month

4. **Expert Judgment**
   - Experienced developers estimate the required effort and cost.
5. **Analogy-Based Estimation**
   - Estimates the new project by comparing it with similar completed projects.

<u>Case study Example : COST ESTIMATION FOR LIBRARY MANAGEMENT SYSTEM</u>

In COCOMO, the project type is selected based mainly on the size, complexity, development environment, and experience of the team.

There are 3 basic project types:

Basic COCOMO Constants for Organic Projects

Parameter	**Constant**	**Value**
Effort coefficient	**a**	**2.4**
Effort exponent	**b**	**1.05**

Parameter	Constant	Value
Development time coefficient	**c**	**2.5**
Development time exponent	**d**	**0.38**

‘

Basic COCOMO Constants for Semi-Detached Projects

Parameter	Symbol	Value
Effort coefficient	**a**	**3.0**
Effort exponent	**b**	**1.12**
Development time coefficient	**c**	**2.5**
Development time exponent	**d**	**0.35**

Basic COCOMO Constants for Embedded Projects

Parameter	Symbol	Value
Effort coefficient	**a**	**3.6**
Effort exponent	**b**	**1.20**
Development time coefficient	**c**	**2.5**
Development time exponent	**d**	**0.32**

Assume the Library Management System has an estimated size of: **KLOC = 10**

Assume it is an **Organic project** because it is a relatively small and simple software system developed by a small team.

Step 1: Calculate Effort

Formula:

Effort = a(KLOC)^b

E = 2.4(KLOC)^{1.05} --values of a and b taken from table

For 10 KLOC:

E = 2.4(10)^{1.05}

E ≈ approx 26.90 person-months

So, the estimated effort is approximately: **26.9 person-months**

Step 2: Calculate Development Time

Formula:

Development Time = c (Effort)^d

T = 2.5(E)^{0.38} --value of c from table

T = 2.5(26.9)^{0.38}

T ≈ approx 8.95 months
Therefore: **Development time ≈ 9 months**

Step 3: Calculate Average Team Size

Team Size = Effort / Development Time = 26.9 / 8.95 ≈ 3 persons

Therefore, approximately **3 people** are required.

Step 4: Calculate Cost

Suppose the average cost of one person-month is **₹50,000**.

Cost = Effort * Cost per Person Month

Cost = 26.9 * ₹50,000 = ₹13,45,000

Final Result

Estimation	Result
Project	Library Management System
Estimated Size	10 KLOC
Project Type	Organic
Effort	26.9 Person-Months
Development Time	8.95 Months
Average Team Size	3 Persons
Cost/Person-Month	₹50,000
Estimated Project Cost	**₹13,45,000**

Conclusion

Using the **Basic COCOMO model**, a 10 KLOC Library Management System is estimated to require approximately **26.9 person-months of effort**, **9 months of development time**, and an estimated development cost of **₹13.45 lakh**, assuming a cost of ₹50,000 per person-month.

Important: The ₹13.45 lakh figure is based on the assumed **10 KLOC** and **₹50,000/person-month**. In an actual project, these values should be replaced with the estimated project size and current team salary/cost.